

DELTA User Guide

Dynamic Energy Load & Trading Analytics Intraday trading optimization dashboard for a 4-plant German utility portfolio on EPEX SPOT (DE-LU bidding zone).

Getting Started

```
pip install -r requirements.txt
streamlit run app.py
```

Open the app in your browser (default: `http://localhost:8501`). Use the sidebar to navigate between pages.

Pages 0-5 work immediately with the included 90-day dataset. The AI Chat page (page 6) requires AWS credentials for Amazon Bedrock AgentCore.

Pages

Home

The landing page shows a real-time snapshot of portfolio health.

What you see: - **5 KPI cards** -- Portfolio P&L, Latest VWAP, TTF Gas price, REMIT Compliance rate, Data Coverage - **2 Critical Findings** -- CCGT Overcommitment Risk (contracts exceed plant capacity) and REMIT Compliance Gap (unreported trades with penalty exposure) - **Architecture overview** -- describes the 5-agent system and analytical capabilities

No interaction needed -- this page loads automatically from the underlying data.

AI Daily Briefing

window.

How to use: 1. **Select a date** using the date picker 2. Click "**Generate AI Briefing**"

What you get: - Price profile chart (Intraday VWAP vs Day-Ahead with CCGT/OCGT SRMC reference lines) - 5 market metrics (Avg VWAP, Peak Price, Min Price, Volatility, Volume) - 6 insight cards: - **Spread Opportunity** -- intraday vs day-ahead arbitrage signals - **Dispatch Economics** -- Clean Spark Spread and profitable dispatch periods - **Renewable Output** -- wind/solar generation and forecast error - **Price Extremes** -- negative price and spike periods - **Trading Performance** -- trade count, P&L, win rate, best trade - **Imbalance Risk** -- system regulation state and penalty spread - **Temperature Impact** warning when CCGT efficiency is derated - **Recommended Actions** -- numbered list of specific trading decisions

Every insight cites its source file and formula.

Portfolio Overview

Full plant fleet and 90-day trading performance with optional filtering.

How to use: - Use the **sidebar filters** to narrow by Strategy (DA_HEDGE, ID_OPTIM, BALANCING, SPREAD) and/or Plant -- all charts and metrics update accordingly

What you see: - **Plant fleet cards** -- 4 plants with capacity, technology, efficiency, and minimum stable load - **90-day KPIs** -- Total P&L, Average Clean Spark Spread, REMIT Compliance Rate, Total Traded Volume - **P&L by Strategy** -- bar chart breaking down profits across DA_HEDGE, ID_OPTIM, BALANCING, and SPREAD strategies - **Trade Count** -- grouped bar showing BUY vs SELL trades per strategy - **Cumulative P&L** -- line chart showing portfolio equity curve over time - **Contract Obligations** -- table of all 6 active contracts with volume, price, tolerance, and penalties - **CCGT Overcommitment Warning** -- flags when peak-hour contracts exceed 430 MW capacity - **Fuel & Carbon Summary** -- average TTF gas, EU ETS carbon, and intraday VWAP with 90-day ranges

Market Analysis

Deep dive into electricity prices, fuel markets, and renewable forecasts.

How to use: 1. **Set the date range** with Start Date and End Date pickers 2. **Switch between 5 tabs:**

Tab	What it shows
Prices	Intraday VWAP vs Day-Ahead with high/low band, 4 summary metrics
Spreads	Intraday-DA spread histogram and daily average spread bar chart
Fuel & Carbon	TTF Gas and EU ETS Carbon dual-axis time series over 90 days
Renewables	Wind or Solar forecast with P10/P90 confidence bands, forecast error distribution. Toggle between Wind and Solar with the radio button
Distribution	VWAP box plot by hour-of-day and price heatmap (hour x day-of-week)

Plant Dispatch

Runs the merit-order dispatch optimizer for a selected day.

How to use: 1. **Select a date** using the date picker 2. Click "**Optimize Dispatch**"

What you get: - **4 summary metrics** -- Total Generation (MWh), Revenue, Cost, and Margin - **Dispatch schedule** -- stacked area chart showing each plant's generation across 96 quarter-hourly periods, overlaid with the market price - **Profit waterfall** -- waterfall chart showing each plant's margin contribution - **Merit-order stack** -- step chart ranking plants by marginal cost (cheapest dispatched first) - **Marginal cost curves** -- CCGT and OCGT cost at varying load levels, showing part-load efficiency impact - **Per-plant summary table** -- generation, revenue, cost, margin, and capacity factor per plant - **Expandable detailed dispatch table** -- every 15-minute period with source citations

The optimizer uses Willans-line part-load efficiency, temperature-corrected capacity derating, and respects minimum stable load, ramp rates, and contract obligations.

Scenario Simulator

What-if analysis comparing base case dispatch against modified market conditions.

How to use: 1. **Select a date** using the date picker 2. **Adjust sidebar sliders:** - **Gas Price** (EUR/MWh) -- shifts CCGT and OCGT SRMC - **Carbon Price** (EUR/tCO₂) -- impacts CO₂ cost component - **Wind Capacity Factor** -- scales wind output (1.0 = actual forecast) - **Solar Capacity Factor** -- scales solar output - **Demand / Obligation Factor** -- scales contract obligations up or down 3. Click "**Run Scenario Comparison**"

What you get: - **4 delta metrics** -- Generation, Revenue, Cost, Margin with change vs base case - **Side-by-side dispatch charts** -- stacked area showing how the generation mix shifts - **Per-plant comparison table** -- base vs scenario generation and margin for each plant - **Textual analysis** -- explains what each parameter change means for dispatch economics - **Sensitivity heatmap** -- 5x5 grid of Gas x Carbon prices showing portfolio margin across 25 dispatch optimizations

Compliance & Risk

Monitors regulatory compliance and portfolio risk exposure across 4 tabs.

Tab 1: REMIT Compliance - Compliance Rate, Missing Reports, Rejected, Pending metrics - REMIT report status pie chart - Reporting summary with trade counts - Rejected REMIT reports table with rejection reasons - Expandable list of trades missing REMIT reports

Tab 2: Contract Obligations - Per-contract cards showing Committed vs Delivered MWh and deviation percentage - Tolerance breach warnings with penalty exposure in EUR - Committed vs Delivered grouped bar chart - CCGT overcommitment warning with shortfall calculation

Tab 3: Imbalance Exposure - Average Long/Short/Spread prices - System regulation state pie chart (SHORT/LONG/BALANCED) - Imbalance price spread histogram - Imbalance settlement prices time series - Top 10 worst imbalance events table

Tab 4: Value-at-Risk - VaR 95%, VaR 99%, CVaR 95%, Worst Day metrics - Daily P&L histogram with VaR threshold lines - Cumulative P&L with drawdown chart - Rolling 7-day VaR with daily P&L scatter (red dots = VaR breaches) - VaR breach count and confidence assessment

AI Chat

Conversational interface powered by 4 specialist Claude agents on Amazon Bedrock AgentCore.

Setup (one-time per session): 1. Open the sidebar 2. Enter your **AWS credentials** (Access Key ID, Secret Access Key, Session Token) 3. Click "**Save Credentials**" 4. Status shows "Connected" when ready

How to use: - Click any of the **6 example questions** to get started, or type your own question - The orchestrator routes your question to the most relevant specialist agent:

Agent	Expertise
Market Analyst	Prices, spreads, fuel trends, renewable forecasts, weather
Dispatch Optimizer	SRMC, merit-order, start-up costs, ramp constraints, contracts
Compliance Officer	REMIT reporting, contract tolerances, penalty exposure
Risk Manager	P&L analysis, imbalance exposure, strategy performance

Grounding policy: Every answer cites the source file, row number, and formula used. If the data doesn't support a claim, the agent says so.

Click "**Reset Conversation**" to clear chat history and start fresh.

Example questions to try: - "What was the single most profitable trade and why?" - "Calculate the break-even electricity price for a CCGT cold start" - "Which trades are missing REMIT reports? What's the penalty exposure?" - "Is the CCGT overcommitted during peak hours? Show the math" - "Compare wind forecast accuracy -- is there a systematic bias?" - "What was the worst imbalance event in the dataset?"

Data Sources

All data is in the `data/` directory (12 CSV files, 4 reference documents):

File	Records	Description
intraday_prices_epex.csv	8,640	Quarter-hourly VWAP, high, low, volume, spreads
day_ahead_prices.csv	2,160	Hourly DA auction prices, buy/sell volumes
fuel_prices.csv	90	Daily TTF gas, EU ETS carbon, coal, Brent oil
trade_blotter.csv	2,689	All executed trades with P&L
remit_transactions.csv	2,232	REMIT transaction reports filed with ACER
renewable_forecast.csv	8,640	Wind/solar forecasts with P10/P90 bands
imbalance_prices.csv	8,640	Quarter-hourly settlement prices
weather_actuals_forecast.csv	8,640	Temperature, wind, solar at 4 locations

plant_portfolio.csv	4	Plant specs, efficiency, ramp rates, start costs
contract_obligations.csv	6	PPAs and bilateral contracts
marginal_cost_curves.csv	varies	Pre-calculated MC at different load levels
grid_constraints.csv	12,960	Cross-border ATC/NTC, congestion, redispatch

Reference documents: EPEX SPOT market rules, CCGT operating manual, REMIT compliance guide, balancing and imbalance settlement framework.

Key Formulas

Formula	Equation
SRMC	Gas_Price / Efficiency + CO2_Price x CO2_Intensity + VOM
Part-load efficiency	$\eta(P) = \eta_{\max} \times (1 - 0.15 \times (1 - P/P_{\max})^2)$
Temperature correction	-0.5% per degree C above 15 degrees C ISO
Clean Spark Spread	Power_Price - Gas/Efficiency - CO2 x CO2_Intensity
VaR (Historical)	5th percentile of daily P&L distribution
CVaR	Expected loss beyond VaR threshold

Portfolio

Plant	Type	Capacity	Efficiency	Min Stable Load
RHEIN_CCGT	Combined Cycle Gas Turbine	430 MW	58.0%	170 MW
NORDSEE_WIND	Offshore Wind	350 MW	--	--
BAYERN_SOLAR	Solar Park	120 MW	--	--

